

Dynamic Systems

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One of the most common ways to solve dynamic models in macroeconomics is to use approximations. This “perturbation” method allows to solve analytically models with two variables and is the way large-scale models are solved numerically (e.g. by the software “Dynare”). Moreover, it allows to plot phase diagrams and impulse response functions to illustrate dynamic properties of the models. In general, this approach consists of several steps:

1. Use the sequential or recursive methods to derive the optimality conditions.
2. Combine the optimality conditions with the resource constraints into a system of dynamic equations $F(X_{t+1}, X_t) = 0$, where X_t is the vector of endogenous variables.
3. Choose a point of approximation and linearize the system around that point to obtain a system of linear equations $f(x_{t+1}, x_t) = 0$, where x_t corresponds to deviations from the point of approximation.
4. Solve the dynamic system using the Blanchard-Kahn method, the method of undetermined coefficients, etc.

The two other lectures discuss the first step, while most of this lecture will focus on how to solve systems of dynamic equations. We start with the simplest examples and move gradually towards more complex systems.

Linear equations Consider a first-order (one lag) deterministic difference equation:

$$x_{t+1} = \lambda x_t,$$

where parameter $\lambda \in \mathbb{R}$ and initial value $x_0 \in \mathbb{R}$ are some constants. Solving this equation means finding the values of x_t in every period t . This can be easily done by iterating the equation backward:

$$x_t = \lambda x_{t-1} = \lambda^2 x_{t-2} = \dots = \lambda^t x_0.$$

This simple example shows that dynamics of x_t depends crucially on the value of λ :

1. Convergence:

- (a) if $|\lambda| < 1$, the system *converges* to zero $x_t \rightarrow 0$,
- (b) if $|\lambda| > 1$, the system *diverges* from zero $x_t \rightarrow \infty$.

2. Monotonicity:

- (a) if $\lambda > 0$, the dynamics is *monotonic*,
- (b) if $\lambda < 0$, the dynamics is *oscillating*.

The **steady state (SS)** of the system is a point $\bar{x}$ such that if we start from it, we stay there forever, i.e. if $x_t = \bar{x}$, then $x_{t+j} = \bar{x}$ for $\forall j > 0$. It follows that the steady state can be found by solving equation $\bar{x} = \lambda\bar{x}$ and depends again on the value of λ .

3. Steady state:

- (a) if $\lambda \neq 1$, the system has a unique steady state $\bar{x} = 0$,
- (b) if $\lambda = 1$, the system has a continuum of steady states $\bar{x} \in \mathbb{R}$.

The steady state is **stable** if $x_t \rightarrow \bar{x}$ as $t \rightarrow \infty$. Thus, steady state $\bar{x} = 0$ is stable iff $|\lambda| < 1$.

How can one solve a higher-order linear equation? For example, consider a second-order difference equation

$$x_{t+1} = \lambda_1 x_t + \lambda_2 x_{t-1}.$$

It turns out, one can reduce the *order* of the system by increasing its *dimensionality*:

$$z_{t+1} = Az_t, \quad \text{where } z_t \equiv \begin{pmatrix} x_t \\ x_{t-1} \end{pmatrix} \text{ and } A \equiv \begin{pmatrix} \lambda_1 & \lambda_2 \\ 1 & 0 \end{pmatrix}.$$

Thus, solving a high-order equation is isomorphic to solving a system of first-order equations – the problem we discuss next.

Linear systems Suppose next that there are two variables (e.g. consumption and savings) and two linear dynamics equations:

$$x_{t+1} = Ax_t, \quad \text{where } x_t \equiv \begin{pmatrix} x_{1t} \\ x_{2t} \end{pmatrix}$$

and A is a 2×2 matrix. Again, the steady state of the system can be found from $\bar{x} = A\bar{x}$. It follows that there is a unique steady state equal $\bar{x} = 0$ if $|A - I| \neq 0$ and continuum of steady states otherwise. Given initial values x_0 , one can iterate the system backward and obtain

$$x_t = A^t x_0.$$

Thus, solving the dynamic system boils down to finding A^t . The latter can be easily calculated when A is a diagonal matrix

$$\begin{pmatrix} a_{11} & 0 \\ 0 & a_{22} \end{pmatrix}^t = \begin{pmatrix} a_{11}^t & 0 \\ 0 & a_{22}^t \end{pmatrix}$$

and the system is uncoupled, i.e. the dynamics of each variable can be analyzed separately from the other one. This is, however, rarely the case in macro models, and finding A^t is more complicated in general case. To make progress, we use the results from linear algebra that effectively transform an arbitrary A into an (almost) diagonal matrix.

We focus here on the simplest case when the eigenvalues of matrix A are real $\lambda_1, \lambda_2 \in \mathbb{R}$ and different $\lambda_1 \neq \lambda_2$.¹ In this case, there exist a diagonal matrix Λ and a matrix Q such that

$$A = Q\Lambda Q^{-1}, \quad \Lambda = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \text{ and } Q = (\nu^1 \ \nu^2),$$

where ν^1 and ν^2 are the eigenvectors corresponding to eigenvalues λ_1 and λ_2 . Using this decomposition, it is easy to solve for A^t :

$$A^t = Q\Lambda Q^{-1} \times Q\Lambda Q^{-1} \times \dots \times Q\Lambda Q^{-1} = Q\Lambda\Lambda \dots \Lambda Q^{-1} = Q\Lambda^t Q^{-1} = Q \begin{pmatrix} \lambda_1^t & 0 \\ 0 & \lambda_2^t \end{pmatrix} Q^{-1}.$$

To illustrate this approach, consider an example with $A = \begin{pmatrix} 1 & 0.5 \\ 1 & 1.5 \end{pmatrix}$ and follow these steps:

1. Find eigenvalues:

$$|A - \lambda I| = \begin{vmatrix} 1 - \lambda & 0.5 \\ 1 & 1.5 - \lambda \end{vmatrix} = (1 - \lambda)(1.5 - \lambda) - 1 \cdot 0.5 = \lambda^2 - 2.5\lambda + 1 = 0$$

The roots of this equation are $\lambda_1 = 2$ and $\lambda_2 = 0.5$.

2. Find eigenvectors and matrix Q :

$$A - \lambda_1 I = \begin{pmatrix} -1 & 0.5 \\ 1 & -0.5 \end{pmatrix} \Rightarrow v^1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$A - \lambda_2 I = \begin{pmatrix} 0.5 & 0.5 \\ 1 & 1 \end{pmatrix} \Rightarrow v^2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

¹The complex and repeated eigenvalues appear routinely in macroeconomic problems and require special treatment. This material, however, goes beyond the scope of this course and can be found in textbook “Discrete Dynamical Systems” by O.Galor.

Thus, the matrix is given by $Q = \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix}$.

3. Find matrix Q^{-1} : applying the formula

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

we get

$$Q^{-1} = \frac{1}{3} \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix}.$$

4. Solve for x_t :

$$\begin{aligned} x_t = Q\Lambda^t Q^{-1}x_0 &= \frac{1}{3} \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} 2^t & 0 \\ 0 & 0.5^t \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} x_{10} \\ x_{20} \end{pmatrix} \\ &= \frac{1}{3} \begin{pmatrix} 2^t(x_{10} + x_{20}) + 0.5^t(2x_{10} - x_{20}) \\ 2^t(2x_{10} + 2x_{20}) + 0.5^t(-2x_{10} + x_{20}) \end{pmatrix} \end{aligned}$$

Of course, one can only guarantee analytical solution for a system of 2 equations. If the number of variables and equations exceeds 2, numerical methods can be used to solve the system.

Phase diagram The dynamics of two-dimensional systems can be illustrated with phase diagrams. Although this also applies to non-linear systems, we focus here on the case of linear dynamic equations. Drawing a phase diagram includes several steps and starts with finding the iso-lines, i.e. the points where one of the variables does not change. To solve for the iso-lines, it is sufficient to impose $x_{1t+1} = x_{1t}$ and $x_{2t+1} = x_{2t}$. The intersection of the two lines corresponds to the steady state of the model as both variables are stationary at that point. Each iso-line also divides the space into an area where $x_{it+1} > x_{it}$ and an area where $x_{it+1} < x_{it}$. Putting arrows allows to see in which direction the system evolves starting from any given point.

Further progress can be made using the following observation: the decomposition of matrix A allows to switch from a dynamic system for x_t to a dynamic system for $y_t \equiv Q^{-1}x_t$

$$x_{t+1} = Ax_t \Leftrightarrow x_{t+1} = Q\Lambda Q^{-1}x_t \Leftrightarrow Q^{-1}x_{t+1} = \Lambda Q^{-1}x_t \Leftrightarrow y_{t+1} = \Lambda y_t.$$

The important advantage of the dynamic system for y_t is that it is decoupled and its transition can be easily analyzed. Effectively, the idea is to go from the original basis to the one give by the eigenvectors $y_0 = Q^{-1}x_0$, then find the dynamics of y_t , and finally to go back to the original basis $x_t = Qy_t$.

Similarly to the case of one difference equation, the dynamics of the system depends on the eigenvalues λ_1 and λ_2 . Each of them determines the dynamics of y_{it} in the decoupled system. The different combinations of paths for y_{1t} and y_{2t} give rise to different dynamics of x_t around the zero steady state:

1. If $|\lambda_1| < 1$ and $|\lambda_2| < 1$, then $x_t \rightarrow 0$, the system is stable and the origin is called a **sink**.
2. If $|\lambda_1| > 1$ and $|\lambda_2| > 1$, then $x_t \rightarrow \infty$, the system is unstable and the origin is a **source**.
3. If $|\lambda_1| > 1$ and $|\lambda_2| < 1$ (or vice versa), then the system is stable along one dimension and unstable along the other and the origin is a **saddle point**.

Blanchard-Kahn method So far, we have assumed that the initial values of the variables x_0 are known and therefore, the dynamic path x_t can be found by iterating the system *backward*. In contrast, the agents in most macroeconomic models are forward-looking, i.e. their actions today depend on what their expectations about the future. This can be easily seen in the saving problem discussed in the previous lecture, where according to the Euler equation, the optimal consumption today depends on agent's expectations about consumption in the next period $u'(C_t) = \beta R u'(C_{t+1})$. In contrast to the accumulated wealth B_t , the initial value of consumption is not given and has to be determined endogenously. In particular, suppose we can reduce the saving problem to a system

$$x_{t+1} = Ax_t, \quad \text{where } x_t = \begin{pmatrix} b_t \\ c_t \end{pmatrix},$$

where b_t and c_t are some transformation of savings and consumption. Decomposing matrix A we get a phase diagram with multiple transition paths, each corresponding to some initial values of b_0 (given) and c_0 (not known). Each of the paths satisfies both the budget constraint and the Euler equation and hence, to choose c_0 and the equilibrium trajectory we need to use another optimality condition – the transversality condition. In “well-behaved models”, the latter usually implies that the optimal trajectory should eventually lead to the steady state. If there is only one such trajectory, then the equilibrium is unique. If the number of such trajectories is greater than one, then there is indeterminacy and multiplicity of equilibria. Finally, if there are no such trajectories, then the model has no solution.

The Blanchard-Kahn principle generalizes this insight. Partition variables into two classes:

1. **state variables** are predetermined at the beginning of a period by the previous history,
2. **control variables** are not predetermined and can be freely adjusted by the agents.

E.g. household wealth B_t is a state variable and the level of consumption C_t is a control variable. Similarly, the capital stock of a firm K_t is a state variable, while the level of investment

I_t is a control.² Denote the number of control variables with m and the number of eigenvalues greater than one (in absolute terms) $|\lambda_i| > 1$ with n . Then the Blanchard-Kahn principle says that the system has (i) exactly one solution if $n = m$, (ii) multiple solutions if $n < m$, (iii) no solution if $n > m$.

As an example consider again the case when $x_t = \begin{pmatrix} b_t \\ c_t \end{pmatrix}$ and $A = \begin{pmatrix} 1 & 0.5 \\ 1 & 1.5 \end{pmatrix}$. As shown above, the eigenvalues of the matrix are $\lambda_1 = 2$ and $\lambda_2 = 0.5$ and hence, $n = m$ and the model has a unique solution. To find it, consider the equation of the decoupled system that corresponds to λ_1 :

$$y_{1t+1} = \lambda_1 y_{1t}.$$

Iterating this equation *forward* we get

$$y_{1t} = \lim_{j \rightarrow \infty} \lambda_1^{-j} y_{1t+j}.$$

Assume that the transversality condition requires convergence to the steady state. Then $\lim_{j \rightarrow \infty} y_{1t+j} = 0$ and $y_{1t} = 0$. Another way to understand this condition is to notice that if $y_{1t} \neq 0$ in some period t , then $y_{1t} \rightarrow \infty$ and the system does not converge to zero. Given $y_t = Q^{-1}x_t$, condition $y_{1t} = 0$ implies $b_t + c_t = 0$. Thus, for any given initial value of b_0 , we can find the initial value of consumption $c_0 = -b_0$ and then iterate the system to solve for the equilibrium path (b_t, c_t) , which corresponds to moving along **saddle path** y_{2t} .

Linearization The equilibrium conditions in macroeconomic models are usually not linear and there are no general mathematical tools that can be used to solve such systems. To make progress, economists often focus on linear approximations to the equilibrium systems, which can then be solved using the techniques discussed above. This approach includes:

1. Choosing a point of approximation. If the economy is stable, then it is reasonable to assume that it spends most of the time in the neighbourhood of the steady state and hence, the latter can be used as a point of approximation.
2. Linearizing the equilibrium conditions. Because the deviations in levels are sensitive to denomination (pounds vs. pence) and hard to interpret, economists usually focus on log-percent deviations from the steady state.

The following approach works in most cases:

$$X_t = \bar{X} \frac{X_t}{\bar{X}} = \bar{X} e^{\log \frac{X_t}{\bar{X}}} \equiv \bar{X} e^{x_t} \approx \bar{X} (1 + x_t),$$

²However, the same macro variables can play the role of states or controls depending on a particular model.

where x_t is the log-deviation of X_t from the steady-state value $\bar{X}$. A few useful properties:

$$\begin{aligned} Z_t = X_t^a &\Rightarrow z_t = ax_t, \\ Z_t = X_t Y_t &\Rightarrow z_t = x_t + y_t, \\ Z_t = X_t + Y_t &\Rightarrow z_t = ax_t + (1-a)y_t, \quad a \equiv \frac{\bar{X}}{\bar{Z}} = 1 - \frac{\bar{Y}}{\bar{Z}}. \end{aligned}$$

Example Consider again the savings model discussed in the previous lecture.³ To simplify the analysis, assume the CRRA preferences, $\beta R = 1$, and any shocks to Y_t are completely unexpected. The equilibrium system includes the household budget constraint

$$B_{t+1} = RB_t + Y_t - C_t,$$

and the Euler equation

$$C_t^{-\sigma} = C_{t+1}^{-\sigma}.$$

Solving the model consists of several steps:

1. Choose the point of approximation: this model does not have a unique steady state, which complicates the first step. Instead, we are going to use the initial level of wealth $\bar{B} = B_0 > 0$. Substitute this value into the budget constraint to find the steady-state value of consumption $\bar{C} = (R-1)\bar{B} + \bar{Y}$. Define the exogenous ratio $\alpha \equiv \frac{\bar{Y}}{\bar{B}}$ and note that $\frac{\bar{C}}{\bar{B}} = R - 1 + \alpha$.
2. Log-linearize the equilibrium system: apply the formulae from above to the EE

$$\bar{C}^{-\sigma}(1 - \sigma c_t) = \bar{C}^{-\sigma}(1 - \sigma c_{t+1})$$

and simplify to

$$c_t = c_{t+1}.$$

Similarly, the budget constraint can be expressed as

$$\bar{B}(1 + b_{t+1}) = R\bar{B}(1 + b_t) + \bar{Y}(1 + y_t) - \bar{C}(1 + c_t),$$

which after simplification becomes

$$b_{t+1} = Rb_t + \frac{\bar{Y}}{\bar{B}}y_t - \frac{\bar{C}}{\bar{B}}c_t,$$

³As shown below, this model is not stationary and therefore, local approximations might be not very accurate. We use this model only for pedagogical reasons to show how it can be solved using different methods.

or equivalently,

$$b_{t+1} = Rb_t + \alpha y_t - (R - 1 + \alpha)c_t,$$

Thus, the linearized system is

$$x_{t+1} = Ax_t + Dy_t, \quad x_t = \begin{pmatrix} b_t \\ c_t \end{pmatrix}, \quad A = \begin{pmatrix} R & -(R - 1 + \alpha) \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad D = \begin{pmatrix} \alpha \\ 0 \end{pmatrix}$$

3. Diagonalize matrix A :

$$|A - \lambda I| = (R - \lambda)(1 - \lambda) = 0$$

The roots of this equation are $\lambda_1 = R > 1$ and $\lambda_2 = 1$. The corresponding eigenvectors are given by

$$A - \lambda_1 I = \begin{pmatrix} 0 & -(R - 1 + \alpha) \\ 0 & 1 - R \end{pmatrix} \Rightarrow v^1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$A - \lambda_2 I = \begin{pmatrix} R - 1 & -(R - 1 + \alpha) \\ 0 & 0 \end{pmatrix} \Rightarrow v^2 = \begin{pmatrix} R - 1 + \alpha \\ R - 1 \end{pmatrix}$$

It follows that

$$\Lambda = \begin{pmatrix} R & 0 \\ 0 & 1 \end{pmatrix},$$

$$Q = \begin{pmatrix} 1 & R - 1 + \alpha \\ 0 & R - 1 \end{pmatrix},$$

$$Q^{-1} = \frac{1}{R - 1} \begin{pmatrix} R - 1 & -(R - 1 + \alpha) \\ 0 & 1 \end{pmatrix}$$

and

$$Q^{-1}D = \begin{pmatrix} \alpha \\ 0 \end{pmatrix}.$$

4. Find the saddle path: define $z_t \equiv Q^{-1}x_t$ and consider the first equation of the system

$$z_{t+1} = \Lambda z_t + Q^{-1}Dy_t$$

that corresponds to the eigenvalue greater than one:

$$z_{1t} = -\frac{\alpha}{R}y_t + \frac{1}{R}z_{1t+1} = \cdots = -\frac{\alpha}{R} \sum_{j=0}^{\infty} R^{-j}y_{t+j} + \lim_{j \rightarrow \infty} R^{-j}z_{1t+j}$$

The last term is equal zero according to the transversality condition. Given that income

shocks are unexpected, we have $y_{t+j} = 0$ for $j > 0$. It follows that $z_{1t} = -\frac{\alpha}{R}y_t$ and substituting in the definition of z_t we get

$$(R-1)b_t - (R-1+\alpha)c_t = -\frac{\alpha}{R}(R-1)y_t.$$

Thus, the saddle path is given by

$$c_t = \frac{R-1}{R-1+\alpha} \left(b_t + \frac{\alpha}{R}y_t \right).$$

5. Find the law of motion for the state variable: substitute the optimal consumption function into the budget constraint:

$$b_{t+1} = b_t + \frac{\alpha}{R}y_t.$$

Thus, the wealth of households b_t follows a random walk with innovations $\frac{\alpha}{R}y_t$.

Continuous time Similar methods can be used to solve dynamic models in continuous-time. The *difference* equations are replaced with the *differential* equations in this case. Given this analogy, we only scetch the key ideas. Start again with a one-dimensional linear equation

$$\dot{x}_t = \lambda x_t,$$

which can also be written as

$$\frac{dx_t}{x_t} = \lambda dt.$$

Integrating the latter expression, we get

$$\log \frac{x_t}{x_0} = \lambda t$$

or equivalently,

$$x_t = x_0 e^{\lambda t}.$$

Indeed, taking a derivative with respect to time, we get the initial equation

$$\frac{dx_t}{dt} = x_0 \lambda e^{\lambda t} = \lambda x_t,$$

confirming the solution. It is clear from this expression that $x_t \rightarrow 0$ if $\lambda < 0$ and $x_t \rightarrow \infty$ if $\lambda > 0$. The steady state can be found from $\dot{x}_t = 0$ and is unique as long as $\lambda \neq 0$. Thus, the threshold for λ of 0 plays the same role in continuous time as the threshold of ± 1 in discrete time. This analogy can be seen clearly if we rewrite the difference equation for an arbitrary

period of time $x_{t+\Delta} = \Lambda x_t = (1 + \lambda\Delta)x_t$ and transform it as $\frac{x_{t+\Delta} - x_t}{\Delta} = \lambda x_t$. Taking the limit $\Delta \rightarrow 0$, we get the differential equation and the mapping $\lambda = \Lambda - 1$.

Consider next the system of linear differential equations

$$\dot{x}_t = Ax_t,$$

where x_t is a vector. If matrix A is diagonal, then the system is decoupled and each equation can be solved independently. Otherwise, we can use the same decomposition $A = Q\Lambda Q^{-1}$ to transform the system into a decoupled system for $y \equiv Q^{-1}x_t$:

$$\dot{y}_t = \Lambda y_t.$$

The Blanchard-Kahn principle still applies: to have a unique solution, the number of control variables in a model has to be equal to the number of eigenvalues greater than one. The continuous-time equations can also be linearized in the same way as the discrete-time analogs.